An Intelligent Computer Vision Framework for Wafer Die Defect Detection, Localization, Classification, Rotation Robustness, and Potential Root-Cause Analysis

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IN ONE SENTENCE We will build and evaluate a die-to-wafer inspection framework that detects and classifies defective dies, localizes defects where annotations permit, tests robustness to rotation and viewing angle, and links wafer-level patterns to *candidate* causes as evidence-graded hypotheses, not verified diagnoses.

1 PROBLEM & MOTIVATION

Yield problems in wafer fabrication first appear as defective dies and as *spatial patterns* of failures across the wafer (edge rings, scratches, local clusters). Manual review is slow and inconsistent, and automated methods often address only one part of the task: image-level defect classification or wafer-map pattern recognition [1–3]. Real data also vary in orientation, alignment and viewing geometry, and unseen defect types appear.

Research question: can one modular framework link die-level evidence to wafer-level patterns reliably under geometric variation, and how far can those patterns support root-cause investigation without over-claiming?

2 RESEARCH OBJECTIVES

- **O1** Detect normal and defective dies.
- **O2** Classify recognizable defect categories; localize defect regions where annotations permit.
- **O3** Analyze wafer-level defect distributions and spatial patterns.
- **O4** Evaluate robustness to rotation, alignment errors and viewing-angle variation.
- **O5** Investigate potential root causes from defect patterns and, where available, process metadata.

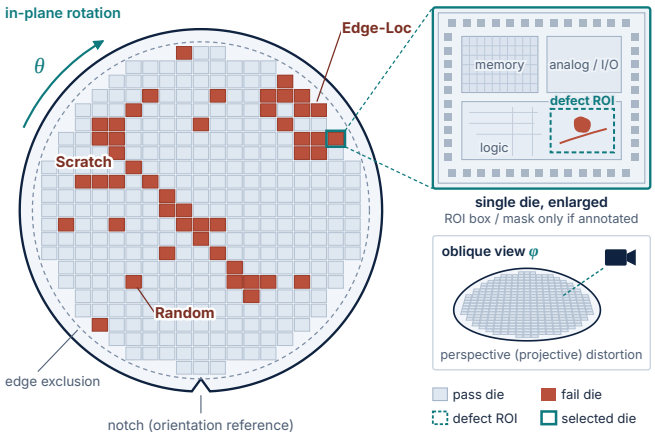


Fig. 1. Schematic, not real data: wafer map with per-die pass/fail states and WM-811K-style patterns [4]; one die enlarged with a localized defect; inset: perspective distortion.

3 PROPOSED METHOD · MODULAR PIPELINE

INPUT

01 Wafer Image / Wafer Map
- Optical / SEM die images
- Wafer maps (pass/fail)
- Process metadata, if any

DIE-LEVEL PERCEPTION

02 O4 Alignment & Preprocessing
- OpenCV: notch and die-grid registration
- Homography correction
- Normalization

03 O1 Die-Level Detection
- Die cropping from grid
- Normal vs. defective: CNN / ResNet
- Imbalance-aware training

04 O2 Classification & Localization
- Defect-type classifiers
- Detection / segmentation if annotated
- Anomaly detection

ROBUSTNESS

05 O4 Rotation-Robustness Evaluation
- Controlled θ, shift, tilt φ
- Augmentation vs. invariant designs
- Score-vs-angle curves

WAFER-LEVEL ANALYTICS

06 O3 Spatial Pattern Analysis
- Wafer-map pattern classes
- Clustering; radial and angular statistics

07 O5 Evidence-Based Root-Cause Investigation
- Pattern ↔ process-step links
- Metadata correlation
- Graded hypotheses

Fig. 2. Planned pipeline; chips show the objective each stage serves. Listed techniques are *candidates* to be compared, not final choices, and none has been implemented yet. Planned metrics: per-class recall and macro-F1 (imbalance), mAP / IoU (localization), AUROC (anomaly detection), and accuracy as a function of θ and φ (robustness).

4 RESEARCH GAP & KEY CHALLENGES

CHALLENGE	PLANNED RESPONSE
Subtle, small defects	High-resolution die crops; multi-scale features
Class imbalance; few labels	Re-weighting, augmentation [3], synthesis [2]
Rotation / alignment variation	Notch-based registration; invariant features
Perspective at oblique angles	Homography rectification; tilt-stratified tests
Localizing defects within dies	Detection or segmentation, only if annotated
Unknown defect categories	Anomaly / open-set detection with abstention
Observed pattern ≠ cause	Evidence grading; physical failure analysis

5 NOVELTY & EXPECTED CONTRIBUTION

- **Die-to-wafer integration:** one framework linking die-level detection and localization to wafer-level spatial analytics.
- **Explicit robustness protocol:** reproducible tests under controlled rotation, misalignment and viewing angle.
- **Open-set awareness:** flag unseen defect types instead of forcing a label.
- **Evidence-graded root-cause output:** every hypothesis states its level of support.

LEVEL 1	LEVEL 2	LEVEL 3
Pattern only: candidate cause	+ process metadata: correlated	+ failure analysis: verified

6 DATA & SCOPE

WM-811K [4]: 811,457 real-world wafer maps; a labeled subset covers patterns such as Center, Donut, Edge-Ring and Scratch. Candidate data for O3 and for rotation tests (O4).

Limitation: WM-811K has pass/fail maps, not microscope images. Surface-defect detection and localization (O1–O2) and physical root-cause analysis (O5) need additional annotated images and process metadata.

7 FEASIBILITY & PLAN

- **Phase 1:** review [1]; WM-811K baselines; robustness protocol.
- **Phase 2:** alignment, die detection, localization on image data.
- **Phase 3:** anomaly detection, spatial analytics, root-cause study.
- **Resources:** public data; Python, OpenCV, PyTorch; GPU [to be confirmed].
- **Main risk:** no image or process data. *Fallback:* O3–O4 on wafer maps.

8 KEY REFERENCES

[1] J. Ma *et al.*, "Review of Wafer Surface Defect Detection Methods," *Electronics*, vol. 12, no. 8, Art. 1787, 2023. doi.org/10.3390/electronics12081787 *Survey; defect causes.*

[2] D. A. Altantawy and M. A. Yakout, "Sparse deep encoded features with enhanced sinogramic red deer optimization for fault inspection in wafer maps," *J. Intell. Manuf.*, 2025 (online 2024). doi.org/10.1007/s10845-024-02377-4 *Wafer-map faults; imbalance handling.*

[3] F. López de la Rosa *et al.*, "Defect detection and classification on semiconductor wafers using two-stage geometric transformation-based data augmentation and SqueezeNet lightweight convolutional neural network," *Comput. Ind. Eng.*, Art. 109549, 2023. doi.org/10.1016/j.cie.2023.109549 *Geometric augmentation; lightweight CNN.*

[4] M.-J. Wu, J.-S. R. Jang and J.-L. Chen, "Wafer map failure pattern recognition and similarity ranking for large-scale data sets," *IEEE Trans. Semicond. Manuf.*, vol. 28, no. 1, pp. 1–12, 2015. doi.org/10.1109/TSM.2014.2364237 *Source of WM-811K.*